

```
import random
```

```
def number_guessing_game():
```

```
# 1. Generate a random number between 1 and  
100
```

```
secret_number = random.randint(1, 100)
```

```
attempts = 0
```

```
print("=== Welcome to the Number Guessing  
Game! ===")
```

```
print("I am thinking of a number between 1 and  
100.")
```

```
# 2. Start the game loop
```

```
while True:
```

```
try:
```

```
# Get the player's guess
```

```
guess = int(input("Enter your guess: "))
```

```
attempts += 1
```

```
# 3. Check the player's guess
```

```
if guess < 1 or guess > 100:
```

```
    print("Please guess a number within the 1 to 100  
    range.")
```

```
elif guess < secret_number:
```

```
    print("Too low! Try a higher number.")
```

```
elif guess > secret_number:
```

```
    print("Too high! Try a lower number.")
```

```
else:
```

```
    print(f"\n🎉 Congratulations! You guessed the  
    number in {attempts} attempts!")
```

```
    break # Exit the loop and end the game
```

```
except ValueError:
```

```
    # Handle non-integer inputs gracefully
```

```
    print("Invalid input! Please enter a valid whole  
    number.")
```

Run the game

```
if name == "main":  
    number_guessing_game()
```